Cockpit Voice Recorder (CVR) – General Data

1. General

The Cockpit Voice Recorder (CVR) is used to record aircraft audio for the purpose of investigating aircraft accidents and incidents. The CVR can record the data for up to 120 minutes.

For aircraft with S.O.O. 6388 incorproated, the Cockpit Voice Recorder (CVR) is not installed.

The Cockpit Voice Recorder (CVR) is installed as part of the IFR Operations Bundle customer Standard Order Option 6209.

For aircraft with S.O.O. 6209 incorporated, the CVR and associated systems may be removed from the aircraft by incorporating customer Standard Order Option Mod S.O.O. 6292.

For aircraft with S.O.O. 6388 incorporated, the CVR and associated systems are installed by incorporating customer Standard Order Option S.O.O. 6384.

2. Description

The Honeywell AR–120 Cockpit Voice Recorder is a four channel CVR with 120 minute recording capacity. The AR–120 uses flash memory modules to record aircraft audio, and has no moving parts. It is equipped with an underwater location beacon.

The Cockpit Voice Recorder is contained in an environmentally protective enclosure. The recorder is located in the rear avionics compartment, at approximately station 415, where the survival rate is high. The recorder has a flange which allows the unit to be mounted directly onto the avionics shelf. 28 Volt DC power is provided from the left bus through a 5 amp CVR circuit breaker on the avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data).

The audio control panels accept various aircraft audio signals, and output all required audio (except the cockpit area microphone) to the CVR via ARINC 429 (refer to 23-53-00, Audio Control Panel – General Data).

The following aircraft systems interface with the CVR system:

A. Cockpit Area Microphone

The cockpit area microphone is mounted in the center of the instrument panel, above the glareshield. The cockpit area microphone is intended to capture the cockpit ambient environment on the CVR. The pre-amplification and conditioning are accomplished within the CVR.

B. Modular Avionics Unit (MAU)

Each channel of the Custom Input/Output Modules (CSIO) outputs alert audio to the audio control panels. This audio is included in the audio panel output to the CVR.

The Generic Input/Output Module (GIO) module transmits Greenwich Mean Time (GMT) to the CVR over A429 (label 150).

C. Impact Switch

The impact switch is an inertia operated device which removes the power to the recorder in the event of a crash, thus keeping the recorded data intact. A deceleration of 3 g or greater will trigger the switch. The impact switch is located in the rear avionics compartment, outboard of the FDR, and is the forward of the two switches. A red lamp on the switch will illuminate when the switch has been tripped.

D. CVR Monitor Jack

A CVR monitor jack is provided on a panel in the rear avionics compartment, bottom shelf, at approximately station 421. The purpose of the jack is to provide a means to test the various audio inputs to the CVR.

E. Underwater Location Beacon

The Underwater Location Beacon (ULB) is a cylindrical transmitter mounted to the front of the crash survivable memory unit. When the sensor comes in contact with water the ULB emits ultrasonic pulses to assist in the location of the submerged recorder.

3. Adjustment/Test

A. Cockpit Voice Recorder

1) If a fault condition is detected by the Built-In Test Equipment (BITE), a caution level CAS message will be displayed. Pulling the CVR (K3) circuit breaker should trigger the CVR Fail CAS.

2) Function Test

a) Gain access to the CVR audio jack on the rear maintenance panel, which is accomplished through the aft baggage compartment, into the rear avionics compartment.

b) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

c) While monitoring the CVR audio jack, do the following and ensure they are heard clearly.

1 Pull both PILOT AUDIO (3E) and COPILOT AUDIO (4E) CB's and speak into the cockpit area mic.

2 Reset the PILOT AUDIO CB (3E) and:

1 Speak into the pilot's headset.

2 Speak into the co-pilot's headset

3 Speak into the observer jack.

4 Speak into the cockpit area mic.

5 Trigger various alert audio sources such as altitude alert, engine fire test, stall warning, etc.

6 Tune a VHF Com radio and do a radio check, ensure both the sidetone and receive audio are heard.

3 Pull PILOT AUDIO CB (3E) and reset the COPILOT AUDIO CB (4E) and:

1 Repeat steps in Para 3.A.2).c).2.

4 Reset the PILOT AUDIO CB (3E).

d) Pull the CVR (K3) circuit breaker and ensure the CVR Fail message is present.

e) Reset circuit breaker and confirm the CAS messages clear.

f) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

B. Underwater Locator Beacon (ULB) – Dukane DK100 and DK120

1) ULB Battery Test

Note Use a high impedance voltmeter (input impedance of 10 M ohms) to do the following procedure.

a) Ensure the ULB is clean and dry, refer to Para 6.A.1).

b) Place the negative lead of the voltmeter on the water switch pin, and the positive lead on the case of the ULB.

c) Read the battery voltage on the voltmeter.

d) Check the ULB label for the battery code. ULBs with codes B, C, or D, have a minimum acceptable voltage of 2.97 volts. If the battery voltage is below minimum acceptable, the battery requires replacement.

2) ULB Operational Test

A Dukane 42A12 Ultrasonic Test Set, or equivalent, is required to do the operational test.

a) Set the TUNING knob between 35 and 40 kHz.

b) Turn the GAIN knob clockwise and check for operation by making a noise (such as jingling keys or coins) near the microphone.

c) Activate the ULB by shorting the water switch pin to the case with a jumper.

d) Adjust the tuning knob, if required, to produce the best audible signal.

e) Proper operation is indicated by a pulsing, audible tone. The ULB should pulse at least 5 times in 10 seconds.

f) Remove the jumper and confirm the pulsing stops.

g) Turn the GAIN knob to OFF position.

4. Removal/Installation

WARNING INSTALLATION OF THIS COMPONENT CONTAINS PROTECTION FEATURES THAT ARE CRITICAL TO PREVENTING DEGRADATION FROM THE EFFECTS OF HIGH INTENSITY RADIATED FIELDS (HIRF) AND LIGHTNING. ENSURE INTEGRITY OF THE HIRF/LIGHTNING PROTECTION FEATURES ARE MAINTAINED IN ACCORDANCE WITH THE FOLLOWING:

CHASSIS GROUNDING – ALL RACK MOUNT/REMOTE MOUNT UNITS SHALL BE ELECTRICALLY BONDED TO THE AIRFRAME. ENSURE THAT THE MATING SURFACES BETWEEN THE LRU MOUNTING TRAY (OR LRU MOUNTING FEET IF A TRAY IS NOT USED) PROVIDE A LOW IMPEDANCE ELECTRICAL PATH. ALL MATING SURFACES MUST BE FREE OF ALL PAINT AND OTHER NONCONDUCTIVE ELEMENTS. REFER TO BONDING REQUIREMENTS IN 20-75-00, ELECTRICAL BONDING AND GROUNDING OF AIRCRAFT STRUCTURES – GENERAL DATA.

SHIELD GROUNDING – VERIFY SHIELD GROUNDING AT CONNECTOR BACKSHELLS AS PER REQUIREMENTS IN PSM 1-64-2W.

BACKSHELLS OR STRAIN RELIEFS – VERIFY PROPER ELECTRICAL BONDING BETWEEN THE BACKSHELL/STRAIN RELIEF AND THE MOUNTING TRAY OR LRU CASE AS PER REQUIREMENTS IN PSM 1-64-2W.

A. CVR

1) Removal of CVR

a) Ensure the aircraft electrical system is OFF.

b) Pull circuit breaker CVR (K3) (Left Bus) on the avionics circuit breaker panel and fit safety clips.

c) Gain access to the CVR through the rear baggage compartment to the aft avionics compartment.

d) Remove (6) mounting screws and washers.

e) Remove unit.

Note Stow connectors of the removed unit.

2) Installation of CVR

The CVR has the same form and fit as the FDR, and will accept the same electrical connector. Ensure the CVR is installed in the correct position.

a) Ensure the aircraft electrical system is OFF.

b) Ensure the CVR (K3) (Left Bus) circuit breaker on the avionics circuit breaker panel is pulled and fitted with safety clips.

c) Position unit onto rack and install (6) mounting screws and washers. Ensure unit is secure.

d) Reconnect electrical connection. Ensure the connection is secure and sound.

e) Close access panel.

f) Remove safety clip and close CVR (K3) circuit breaker on the avionics circuit breaker panel.

g) Do a function test, refer to Para 3.A.2).

B. Cockpit Area Microphone

1) Removal of Cockpit Area Microphone

a) Ensure the aircraft electrical system is OFF.

b) Open CVR (K3) circuit breaker on the avionics circuit breaker panel and fit safety clip.

c) Loosen the (2) hex headset screws.

d) Remove the area microphone from the mount to gain access to the wire pigtail and connector.

e) Disconnect the electrical connector.

Note Stow the connectors as required.

2) Installation of Cockpit Area Microphone

a) Ensure the aircraft electrical system is OFF.

b) Ensure the CVR (K3) circuit breaker on the avionics circuit breaker panel is opened and fitted with safety clips.

c) Connect electrical connector. Ensure connection is secure and sound.

d) Position the area microphone on the mount.

e) Tighten the (2) hex headset screws.

f) Remove safety clip and close CVR (K3) circuit breaker on the avionics circuit breaker panel.

g) Do a function test, refer to Para 3.A.2). Only the cockpit area mic portion of Para 3.A.2).c) needs to be completed.

C. Impact Switch

1) Removal of Impact Switch

a) Ensure the aircraft electrical system is OFF.

b) Open CVR (K3) circuit breaker on the avionics circuit breaker panel and fit safety clip.

c) Gain access to the impact switch, which is accomplished through the aft baggage compartment, into the rear avionics compartment.

d) Disconnect electrical connector.

e) Remove (4) mounting screws and washers.

f) Remove unit.

Note Stow the connectors as required.

2) Installation of Impact Switch

a) Ensure the aircraft electrical system is OFF.

b) Ensure the CVR (K3) circuit breaker on the avionics circuit breaker panel is opened and fitted with safety clip.

c) Set unit in position and install (4) mounting screws and washers. Ensure unit is secure.

d) Reconnect electrical connector. Ensure the connection is secure and sound.

e) Close access panel.

f) Remove safety clip and close CVR (K3) circuit breaker on the avionics circuit breaker panel.

g) Do a function test, refer to Para 3.A.2), Para 3.A.2).c) may be omitted for the purposes of this test.

5. Troubleshooting

A. CVR System

1) If both the CVR and FDR Fail messages are present, check that the FDR and CVR are installed in the correct locations.

2) Ensure the CVR (K3) circuit breaker is closed.

3) Ensure the CVR impact switch (forward of FDR impact switch) has not tripped.

4) Ensure the audio panels are installed and operational.

5) Visually inspect CVR connector for damaged wiring or contacts. Ensure electrical connector on CVR is fully seated and locked.

6) If troubleshooting at a wire level is required, refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

7) If repair or service is required, send to appropriately rated avionics repair facility.

6. Servicing

A. Underwater Location Beacon

1) The case and water switch should be cleaned when dust and dirt become apparent. The switch should be cleaned using a soft cloth and mild detergent, then dried with a clean cloth.

2) Change the battery on any of the following conditions:

a) After use in an emergency.

b) If the battery fails the periodic test.

c) On or before the battery replacement date entered on the case of the beacon.

d) There is any evidence of corrosion or leakage.

3) Operation checks of the ULB must be carried out upon installation in accordance with PSM 1–6–7.